

Table 1: **Parameterization of input variables in the PROSPECT-D and 4SAIL models across different parametrizations used in this study.** Abbreviations: $N(\mu, \sigma)|_{min}^{max}$ – truncated normal distribution, $U(a, b)$ – uniform distribution, $C(x)$ – constant value. Minimum and maximum bounds are indicated as subscripts and superscripts.

Parameter	Abbr.	Unit	SNAP	Foliar Codis.	Estevez et al.	Kovacs et al.	Wan et al.
Leaf variables: PROSPECT-D							
Leaf chlorophyll content	CHL	$\mu\text{g cm}^{-2}$	$N(35, 30) _5^{75}$	Gaussian Copula	$N(35, 30) _5^{75}$	$U(0, 100)$	$N(50, 20) _5^{100}$
Leaf carotenoid content	CAR	$\mu\text{g cm}^{-2}$	CHL / 7	Gaussian Copula	CHL / 7	$N(10, 10) _0^{20}$	$N(7.5, 3.5) _0^{15}$
Leaf mass per area	LMA	g cm^{-2}	$N(0.005, 0.001) _{0.001}^{0.03}$	Gaussian Copula	$N(0.005, 0.001) _{0.001}^{0.03}$	$N(0.005, 0.003) _{0.002}^{0.02}$	$N(0.015, 0.005) _{0.001}^{0.03}$
Equivalent water thickness	EWT	cm	NA	Gaussian Copula	$N(0.02, 0.01) _{0.002}^{0.05}$	$N(0.012, 0.006) _{0.005}^{0.035}$	NA
Relative water content	Cw_rel	–	$N(0.75, 0.08) _{0.6}^{0.85}$	NA	NA	NA	$U(0.55, 0.95)$
Leaf anthocyanin content	ANT	$\mu\text{g cm}^{-2}$	$C(0)$	$C(0)$	$C(0)$	$C(0)$	$C(0)$
Leaf structure parameter	N	–	$N(1.5, 0.03) _{1.2}^{1.8}$	$N(1.5, 0.03) _{1.2}^{1.8}$	$U(1.3, 2.5)$	$N(1.5, 0.05) _{1.0}^{2.7}$	$N(1.75, 0.5) _{1.0}^{2.5}$
Fraction of senescent leaves	BROWN	–	$N(0.0, 0.3) _0^2$	$N(0.0, 0.3) _0^2$	$C(0)$	$C(0)$	$U(0.0, 0.2)$
Canopy variables: 4SAIL							
Leaf area index	LAI	$\text{m}^2 \text{m}^{-2}$	OPTUNA	OPTUNA	OPTUNA	OPTUNA	OPTUNA
Hot spot parameter	q	m m^{-1}	$N(0.2, 0.5) _{0.1}^{0.5}$	$N(0.2, 0.5) _{0.1}^{0.5}$	$C(0.01)$	$C(0.01)$	$C(0.01)$
Average leaf angle	LIDFa	°	$N(60, 30) _{30}^{80}$	$N(60, 30) _{30}^{80}$	$U(40, 70)$	$U(40, 70)$	$N(55, 10) _{30}^{80}$
Leaf distribution function	Type-LIDF	–	2	2	2	2	2
Observation / Illumination conditions: 4SAIL							
Linear combination of dark and bright soil spectra	psoil	–	$U(0.0, 1.0)$	$N(0.25, 0.6) _{0.0}^{1.0}$	$U(0.0, 1.0)$	$U(0.0, 1.0)$	$U(0.0, 1.0)$
Sun zenith angle	tts	°	GEE S2 metadata	GEE S2 metadata	GEE S2 metadata	GEE S2 metadata	$U(15, 80)$
View zenith angle	tto	°	GEE S2 metadata	GEE S2 metadata	$C(0)$	$U(-10, 10)$	$U(0, 12)$
Relative azimuth angle	psi	°	GEE S2 metadata	GEE S2 metadata	$C(0)$	$C(180)$	$U(0, 180)$